

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)	(i)		$\text{C}_{12}\text{H}_{22}\text{O}_{11} + 18 [\text{O}] \rightarrow 6 (\text{COOH})_2 + 5 \text{H}_2\text{O}$	1			1		
		(ii)		Mass of the anhydrous acid = 4.05g $\therefore$ Moles of the anhydrous acid = $4.05 / 90 = 0.045$ (1) Mass of the water lost = 1.62g $\therefore$ Moles of water = $1.62/18.02 = 0.090$ (1) Mole ratio acid : water is 1 :2 Value of x is 2 (1) If no working is shown award (1) for the correct answer		3		3	2	
	(b)			Sample of potassium methanoate not pure / not all HCOOK reacted (1) Inadequate heating / not heated for long enough / not heated at a high enough temperature (1)			2	2		2
	(c)	(i)		Allow to settle / test the filtrate (1) Add a few drops of calcium chloride solution and see if a precipitate forms / cloudiness (1)			2	2		2
		(ii)		Moles of calcium oxalate = $2.49/128 = 0.0195$ (1) $\therefore$ Number of moles of potassium oxalate is also 0.0195 Mass of potassium oxalate is $0.0195 \times 166 = 3.24 \text{ g}$ (1) $\therefore$ % of potassium oxalate in mixture = $3.24 \times 100 / 4.69 = 69.1$ (to 3 sig. figs.) (1) (accept values from 68.9 to 69.1 depending on use of significant figures during the calculation) ecf possible		3		3	1	
				Question 9 total	1	6	4	11	3	4